Assignment 01 (R), Sem 2520

Table of contents

Introduction . . . 1
Bayesian inference . . . 1
Check! . . . 4

Introduction

When performing data analysis, a large challenge is always with translation of mathematical set-up to the code. The cookie-cutter analyses are easier to run nowadays, especially with the advent of LLMs. In this assignment, one of the goals is to get you to practice this translation: implementing an analyses we have not seen before. Along the way, I hope you will get practice with reading documenation in R. If you have doubts, or cannot follow certain steps, please do not hesitate to check in with me.

Bayesian inference

In tutorial 2, we observed that Y follows a truncated Poisson with parameter λ if

$$P(Y = y) = P(X = y \,|\, X > 0), \quad y = 1, 2, \ldots$$

where $X \sim Pois(\lambda)$. Given observations $y_1, \ldots, y_n$, the Maximum Likelihood Estimate $\hat{\lambda}$ is given by the solution to

$$\bar{y} = \frac{\hat{\lambda}}{1 - \exp(-\hat{\lambda})}$$

Suppose we observe the following readings from 30 observations of Y:

	$Y=1$	$Y=2$	$Y=3$	$Y=5$
count:	12	14	3	1

1. Create an R function named `f` that implements the following function:

$$f(\lambda, \mathbf{y}) = \bar{y} - \frac{\hat{\lambda}}{1 - \exp(-\hat{\lambda})}$$

   Here is an example of how it works:

```
y_vals <- rep(c(1, 2, 3, 5), times=c(12, 14, 3, 1))
f(2.3, y_vals)
```

```
[1] -0.7562908
```

2. In R, the `uniroot()` function can be used to solve for the zero of a function. Use it to solve for the value of λ that returns 0, for the data above. Extract the root from the output object and save it as `mle_lambda`.

3. We now turn to a Bayesian analysis of the problem. In a Bayesian set-up, the unknown parameter is treated as a random variable. In this case:

$$Y|\lambda \sim \text{truncated Poisson} \tag{1}$$
$$\lambda|\alpha, \beta \sim \Gamma(\alpha, \beta) \tag{2}$$

   The assumed Gamma[1] distribution for λ is referred to as the *prior* distribution. Domain knowledge and subjective beliefs can be captured through this prior, by deciding on α (shape parameter) and β (scale parameter) to be used. But once data is observed, Bayesian inference relies on the *posterior* distribution of λ **conditional on the data**. In this case, it can be shown that

$$\pi(\lambda \mid \alpha, \beta, y_1, \ldots, y_n) = \frac{P(Y_1 = y_1, \ldots, Y_n = y_n|\lambda) f(\lambda|\alpha, \beta)}{\int P(Y_1 = y_1, \ldots, Y_n = y_n|\lambda) f(\lambda|\alpha, \beta) d\lambda} \tag{3}$$

   where

$$P(Y_1 = y_1, \ldots, Y_n = y_n|\lambda) f(\lambda|\alpha, \beta) = \frac{1}{\left(\prod_{i=1}^n y_i!\right) \Gamma(\alpha) \beta^\alpha} \frac{\lambda^{\sum y_i + \alpha - 1} e^{-\lambda(n + 1/\beta)}}{(1 - e^{-\lambda})^n} \tag{4}$$

   Take note that, once the data is observed and a decision has been made on the values of α and β to use, the integrand in the denominator is only a a function of λ. The following code defines the integrand function and computes the integral. Extract the value of the integral and store it as `K`.

[1] More information about the Gamma pdf.

```r
alpha <- 0.001
beta <- 1000

integrand_fn <- function(lambda, y, alpha, beta) {
  n <- length(y)
  log_value <-  (sum(y) + alpha - 1)*log(lambda) - lambda*(n + 1/beta) -
                n*log(1-exp(-lambda))
  exp(log_value)
}
g <- Vectorize(integrand_fn, vectorize.args = c("lambda"))
integrate(g, 0, Inf, y=y_vals, alpha=alpha, beta=beta, abs.tol = 1.0e-16)
```

4. Using the following sequence of λ values, compute the values for the posterior pdf and re-create the following plot.

```r
lam_val <- seq(0, 3, by=0.001)
```

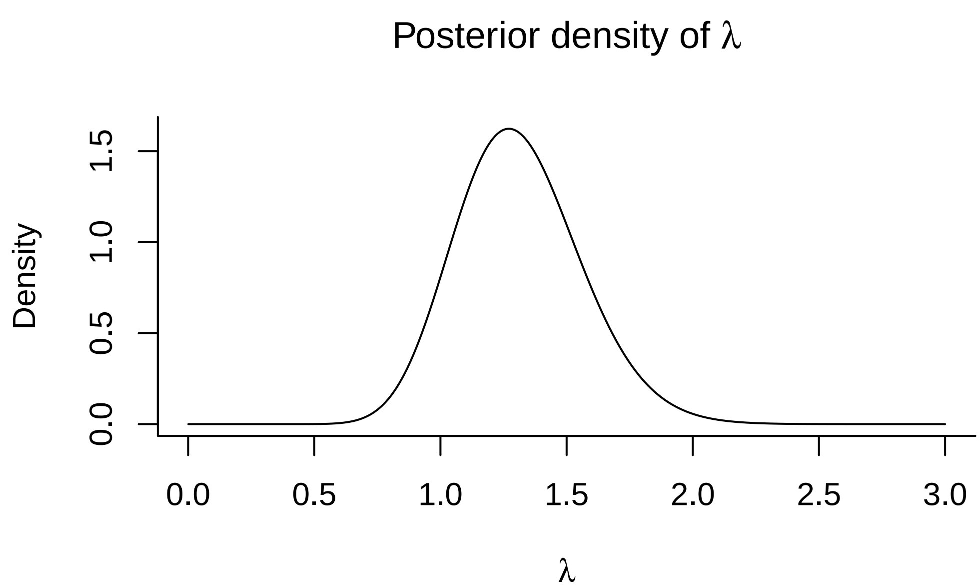

Figure 1: Posterior density

5. Using the computed numeric values for the posterior pdf, estimate the posterior mode. This is an alternative estimator to the MLE.

Check!

Before you submit, ensure that your code creates the following objects in R:

1. `f()`: a function that takes in two arguments, `lambda` and `y_vals`.
2. `mle_lambda`: a scalar (i.e. a numeric vector of length 1)
3. `K`: a scalar (i.e. a numeric vector of length 1)
4. `posterior_mode_lambda`: a scalar (i.e. a numeric vector of length 1).

Also make sure that your code generates the plot above!